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AI for Bharat Hackathon

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Team Name : Bharat CareLink

Team Leader Name : Nitin Bhausaheb Kothwal

Problem Statement : **Bharat CareLink -** AI-Driven Rural Healthcare Referral System

Bharat CareLink is a **cloud-native, AI-driven platform** that revolutionizes how rural healthcare referrals are managed.

> It replaces fragmented paper trails with a digital ecosystem that prioritizes care for those who need it most.

By integrating advanced AWS AI services, we enable real-time risk assessment and ensuring the **highest-risk patients are treated first** at Primary Health Centres (PHCs).

Voice-to-Form

Uses **Amazon Transcribe** for multi-lingual data entry, overcoming literacy barriers for ASHA workers.

Automated Risk Scoring

Leverages **Amazon SageMaker** to classify patients as LOW, MEDIUM, or HIGH risk instantly.

Smart Queues

Prioritized dashboards at PHCs ensure critical cases skip the line for immediate attention.

Real-time Trends

Supervisors monitor district-level health data via live dashboards for proactive intervention.

HOW IS IT DIFFERENT?

EXISTING IDEAS

Traditional digital forms create static queues with simple data entry.

BHARAT CARELINK

"AI-First" Platform

Provides automated medical triage rather than just digitizing paper.

HOW IT SOLVES THE PROBLEM

Offline-First Sync

Eliminates manual delays even in low-connectivity areas.

Automated SLA Monitoring

Ensures follow-up tracking so no patient is left behind.

GenAI Summaries

Distills complex records into actionable summaries for doctors.

KEY USPS

Intelligent Triage

ML-driven risk scores (0.00–1.00) that rank patients by medical urgency.

Voice-to-Form

Multi-lingual voice transcription reduces ASHA worker burnout.

Automated Summarization

Instantly generates clinical case summaries using Amazon Bedrock.

Structured Voice Intake
Digital patient intake with validation and multi-lingual voice support via Amazon Transcribe.

AI Risk Scoring
Real-time ML prediction for Pregnancy, Malnutrition, and TB using Amazon SageMaker models.

PHC Smart Queue
Dynamic dashboard sorted by medical risk score (High to Low) and time urgency.

Closed-Loop Tracking
End-to-end monitoring from ASHA submission to final treatment at PHC.

Automated Alerts
SMS/Email escalations for missed follow-ups triggered via Amazon SNS.

Offline-First Support
Local caching and background synchronization for seamless operation in low-connectivity areas.

1

ASHA Mobile App

Captures patient data via voice & text input.

2

Risk Assessment

SageMaker calculates risk score based on 23+ features.

3

Case Summary

Amazon Bedrock generates clinical summary for doctors.

4

PHC Queue

Referral appears in prioritized doctor's dashboard.

5

Closed Loop

Status updates trigger notifications back to ASHA.

aws Powered by AWS Cloud • Real-time Sync • AI Inference

FRONTEND

- **React Native** — ASHA Mobile App
- **React.js** — Web Dashboard

COMPUTE

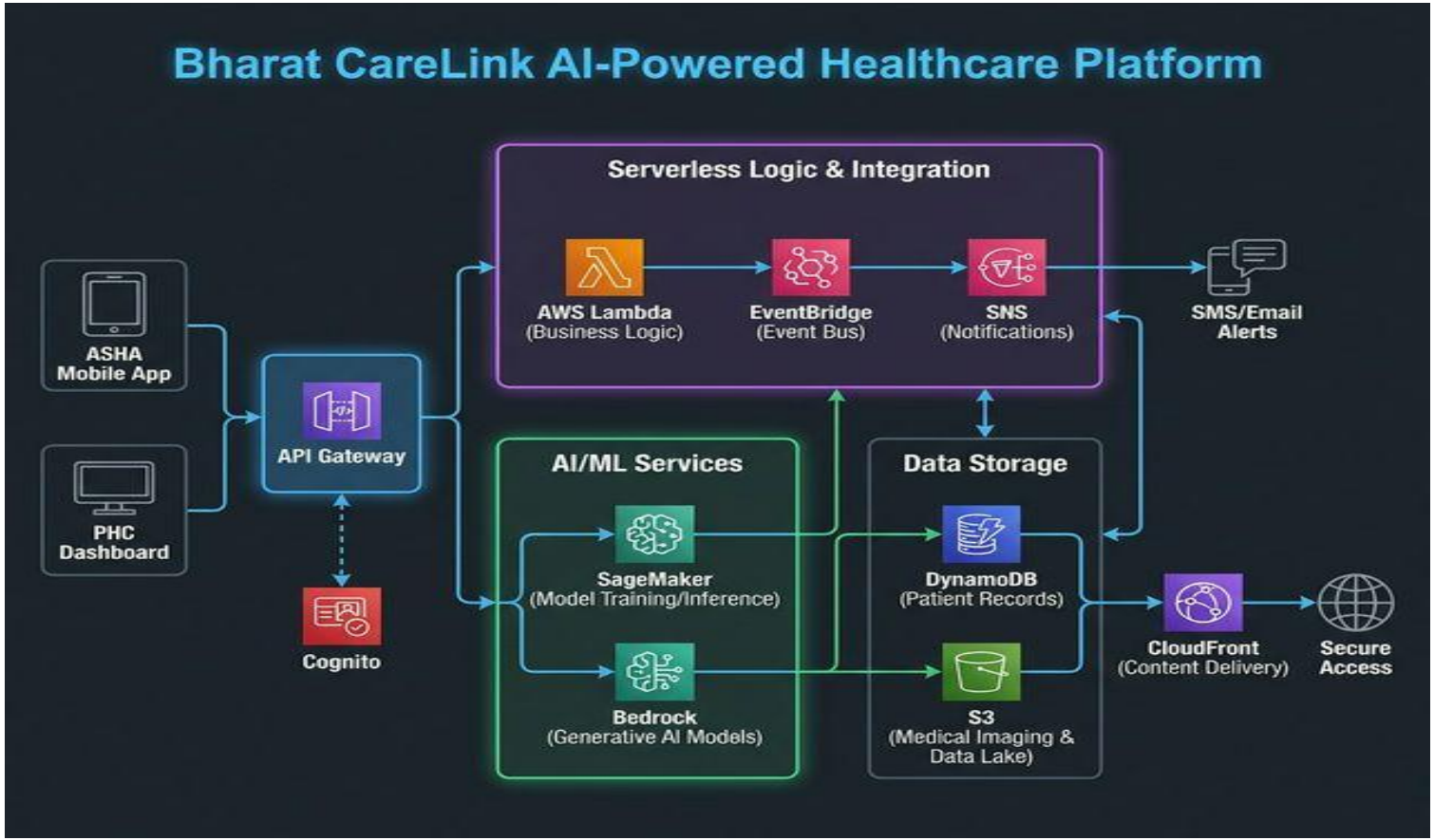
- **API Gateway** — Secure REST APIs
- **AWS Lambda** — Serverless Logic

AI SERVICES

- **Amazon Bedrock** — GenAI Summaries
- **Amazon SageMaker** — ML Risk Models
- **Amazon Transcribe** — Voice-to-Text

DATA STORAGE

- **Amazon DynamoDB** — NoSQL Database
- **Amazon S3** — Object Storage


Bharat CareLink AI-Powered Healthcare Platform
Serverless Logic & Integration
AWS Lambda
(Business Logic)
EventBridge
(Event Bus)
SNS
(Notifications)
SMS/Email
Alerts
ASHA
Mobile App
PHC
Dashboard
API Gateway
Cognito
AI/ML Services
SageMaker
(Model Training/Inference)
Bedrock
(Generative AI Models)
Data Storage
DynamoDB
(Patient Records)
S3
(Medical Imaging &
Data Lake)
CloudFront
(Content Delivery)
Secure
Access

Technologies to be used in the solution

Generative AI
Advanced AI for summarization and insights.

Amazon Bedrock | Claude 3.5 Sonnet

Machine Learning
Core engines for risk scoring and voice processing.

Amazon SageMaker | Amazon Transcribe | AWS Translate

Backend & Events
Serverless architecture for scalable logic.

AWS Lambda | Amazon API Gateway | Amazon EventBridge

Data Storage
Secure and scalable data persistence.

Amazon DynamoDB (NoSQL) | Amazon S3 (Object Store)

Security & Identity
Enterprise-grade protection and access control.

AWS Cognito (RBAC) | AWS KMS (Encryption)

Frontend Application
User interfaces for ASHA workers and doctors.

React Native (Mobile) | React.js (Web Dashboard)

TOTAL MONTHLY COST

~$6,600

Estimated for 50,000 users

Production Scale

FinOps Strategy

Serverless First Approach

Architecture scales down to zero when idle. You only pay for what you use, ensuring minimal costs during off-peak hours in rural PHCs.

Major Cost Drivers

Amazon SageMaker	ML Instances for Training & Inference	~$2,500 per month
Amazon SNS	SMS Notifications for Alerts & Follow-ups	~$1,500 per month
Amazon Transcribe	Multi-lingual Voice Processing	~$750 per month
Other Services	Lambda, DynamoDB, S3, API Gateway	~$1,850 per month

Roadmap & Future Vision

Phased Rollout Strategy & Success Metrics

Phase 1: Pilot
0–6 Mo

- Deploy in 50 PHCs
- Fine-tune Risk Models
- Gather ASHA Feedback

Phase 2: Scale
6–18 Mo

- Add 5 Regional Languages
- Integrate State Registries
- Expand to District Level

Phase 3: National
18+ Mo

- ABHA ID Integration
- Federated Learning
- Full Data Privacy Compliance

Key Success Metrics
Targeted Impact

30%
Reduction in Referral Delays

80%
Precision in High-Risk Classification

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Thank You